

PENDYALA SIRI

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PROFESSIONAL SUMMARY

AI & Data Science graduate (2026) with 10+ months of production experience and a first-author publication on LLM hallucination detection. Supports business-critical applications, PostgreSQL databases and AI services in a secure, restricted-access production environment — spanning application support and system administration, REST API integrations, batch job processing, release management and automated testing. AI-first by default: uses Claude and ChatGPT daily for development, debugging and documentation, with strong ownership and cross-functional collaboration.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, JavaScript, TypeScript, Java, R, Bash | PostgreSQL, MongoDB, SQLite, Qdrant; data modeling, SQL query optimization, schema migrations (Alembic)
APIs, Integrations & Web: REST API design, FastAPI, Express.js, WebSockets, JWT authentication, system integration; React, Next.js, Tailwind CSS
Application Support & Administration: Production support, incident troubleshooting & root-cause analysis, service configuration, user access management (RBAC), release & deployment management, runbooks & documentation
Batch Processing & Automation: Celery, RabbitMQ, asynchronous/background job workflows, scheduled jobs, GitHub Actions (CI/CD), workflow automation
Testing & QA: pytest, Playwright, unit / integration / regression testing, edge-case reproduction, defect triage with developers and QA
DevOps & Infrastructure: Docker, K3s (Kubernetes), HAProxy, Linux, Git, Nexus Repository, MinIO (S3-compatible storage), NVIDIA Triton
AI & Productivity Tools: Claude, ChatGPT, Codex; PyTorch, computer vision, RAG, prompt engineering, model optimization
Business & Functional: Requirements documentation, business process analysis, functional specifications, stakeholder communication, MS Excel

PROFESSIONAL EXPERIENCE

Junior Software Engineer

Jul 2026 – Present

Center for Visual Information Technology (CVIT), IIIT Hyderabad

Hyderabad, India

- Administer and support a production AI-driven platform (New Anubhava Mantapa) across application services, databases, AI inference pipelines and infrastructure, within a secure air-gapped network under strict access and change controls.
- Troubleshoot and resolve production application issues across containerized services (Docker, K3s) — root-cause analysis, service configuration changes, and coordinating fixes with developers and testers.
- Manage batch and asynchronous job-processing workflows using Celery and RabbitMQ, integrating AI workloads (NVIDIA Triton, MinIO object storage, Qdrant) with application services for reliable high-volume background processing.
- Administer PostgreSQL databases and schema migrations using Alembic, and configure HAProxy for database and application traffic management, improving production stability and load distribution.
- Support release management — versioned deployments, configuration updates, rollback readiness — govern controlled package distribution via Nexus Repository, and maintain architecture and troubleshooting docs.

AI Engineer Intern

Apr 2026 – Jul 2026

Zeto.studio

Hyderabad, India

- Engineered a low-latency real-time speech streaming pipeline (RTF 0.6, TTFB 1.2s) on a WebSocket-based full-duplex STT–TTS architecture with sentence-level chunk delivery, containerized with Docker.
- Cut per-request latency and output errors by diagnosing bottlenecks through systematic profiling, dynamic batching, model quantization and voice activity detection (VAD).

Junior Software Engineer Intern

Nov 2025 – Apr 2026

Center for Visual Information Technology (CVIT), IIIT Hyderabad

Hyderabad, India

- Designed and deployed an AI-driven Text-to-Speech system (INDIC F5) as a FastAPI REST service for real-time inference, plus computer vision pipelines for automated background removal using alpha-channel masking and chroma-keying.
- Built pytest unit and integration suites covering edge-case failures for regression coverage, and authored architecture, functional specification and troubleshooting documentation.

PROJECTS

InsightHub — Business Intelligence Platform | FastAPI, PostgreSQL, SQL

Reporting platform using multi-table SQL joins, indexed queries and parallel aggregations, with role-based access control for secure multi-user reporting.

KALANETRA — Mohiniyattam Mudra Detection | Python, MediaPipe, scikit-learn

Curated a 300-image dataset across 24 mudra classes and trained a landmark-based classifier reaching 85% accuracy under limited training data.

FitStyle — Body-Aware Fashion Recommendation | Python, KNN, OpenPose

Recommendation system extracting body keypoints via OpenPose and matching similar body types with K-Nearest Neighbours.

EDUCATION

Methodist College of Engineering and Technology — B.E. in Artificial Intelligence and Data Science Nov 2022 – June 2026

CGPA: 8.7 / 10 | Hyderabad, India

RESEARCH & LEADERSHIP

First Author — “Detecting Hallucinations in Large Language Models”, GRENZE International Journal of Engineering and Technology, 2025.

thegrenze.com/public/index.php/abstract/journal/5202

Divisional President (2023–2024) and **Vice President – Education (2022–2023)**, Street Cause MCET; **Director of Education, Hyderabad Youth Assembly Season 12 (2022–2023)**; **Student Ambassador, WE Hub** and **IEEE Student Branch Member (2025–2026)**.